RACE Regular Analysis Through Continuous Exercise

Electrostatics : RACE-01

1. A 10 μC charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are :

(A) 9 μC, 1 μC (B) 5 μC, 5 μC (C) 7 μC, 3 μC (D) 8 μC, 2 μC

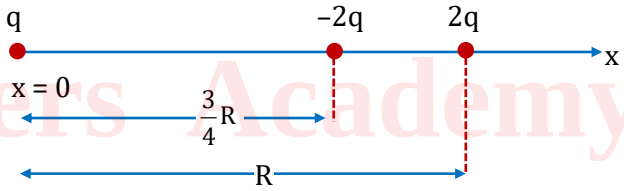
PEO313

2. Three charges $+Q$, q, $+Q$ are placed respectively, at distance, 0, $d/2$ and d from the origin, on the x-axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is :

(A) $+Q/2$ (B) $-Q/2$ (C) $-Q/4$ (D) $+Q/4$

PEO314

3. Three point charges q, $-2q$ and $2q$ are placed on x-axis at a distance $x = 0$, $x = \frac{3}{4}R$ and $x = R$ respectively from origin as shown. If $q = 2 \times 10^{-6}\,C$ and $R = 2\,cm$, the magnitude of net force experienced by the charge $-2q$ is _____N.



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PEO315

4. An infinite number of point charges, each carrying $1\mu C$ charge, are placed along the y-axis at $y = 1m$, $2m$, $4m$, $8m$ _____. The total force on a $1C$ point charge, placed at the origin, is $x \times 10^3\,N$. The value of x, to the nearest integer, is_____.

[Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\,Nm^2/C^2$]

PEO316

5. Force between two point charges q_1 and q_2 placed in vacuum at 'r' cm apart is F. Force between them when placed in a medium having dielectric $K = 5$ at '$r/5$' cm apart will be:

(A) $F/25$ (B) $5F$ (C) $F/5$ (D) $25F$

PEO317

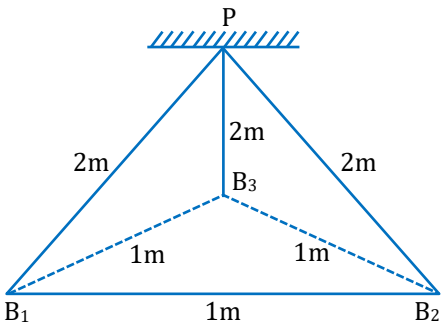
6. If two charges q_1 and q_2 are separated with distance 'd' and placed in a medium of dielectric constant K. What will be the equivalent distance between charges in air for the same electrostatic force ?

(A) $d\sqrt{k}$ (B) $k\sqrt{d}$ (C) $1.5d\sqrt{k}$ (D) $2d\sqrt{k}$

PEO318

For More Join: @IITJEE_Advanced

7. Three identical charged balls each of charge $2C$ are suspended from a common point P by silk threads of $2m$ each (as shown in figure). They form an equilateral triangle of side $1m$. The ratio of net force on a charged ball to the force between any two charged balls will be :



(A) 1 : 1 (B) 1 : 4 (C) $\sqrt{3}:2$ (D) $\sqrt{3}:1$

PEO319

8. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle θ with each other. When suspended in water the angle remains the same. If density of the material of the sphere is 1.5 g/cc, the dielectric constant of water will be _____
(Take density of water = 1 g/cc)

PEO320

9. Two equal positive point charges are separated by a distance $2a$**.** The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes maximum is $\frac{a}{\sqrt{x}}$. The value of x is _____.

Rankers' Academy JEE

PEO321

10. A thin metallic wire having cross sectional area of $10^{-4}\ m^2$ is used to make a ring of radius 30 cm. A positive charge of $2\pi\ C$ is uniformly distributed over the ring, while another positive charge of 30 pC is kept at the centre of the ring. The tension in the ring is _____ N; provided that the ring does not get deformed (neglect the influence of gravity). (given, $\frac{1}{4\pi\epsilon_0}=9\times10^9$ SI units)

PEO322

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	5440	12	B	A	D	3	2	3

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Electrostatics : RACE-02

1. Two charges q and $3q$ are separated by a distance 'r' in air. At a distance x from charge q, the resultant electric field is zero. The value of x is :

(A) $\frac{(1+\sqrt{3})}{r}$ (B) $\frac{r}{3(1+\sqrt{3})}$ (C) $\frac{r}{(1+\sqrt{3})}$ (D) $r(1+\sqrt{3})$

PEO323

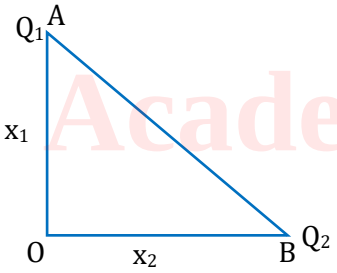
2. Two point charges $q_1(\sqrt{10}\,\mu C)$ and $q_2(-25\,\mu C)$ are placed on the x-axis at x = 1 m and x = 4 m respectively. The electric field (in V/m) at a point y = 3 m on y-axis is, $\left[\text{take}\,\frac{1}{4\pi\varepsilon_0}=9\times10^9\,Nm^2C^{-2}\right]$

(A) $(-63\hat{i}+27\hat{j})\times10^2$ (B) $(81\hat{i}-81\hat{j})\times10^2$

(C) $(63\hat{i}-27\hat{j})\times10^2$ (D) $(-81\hat{i}+81\hat{j})\times10^2$

PEO324

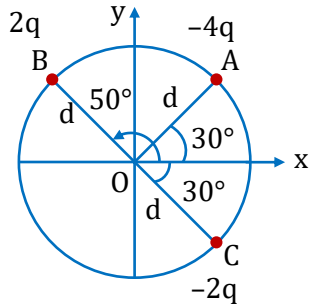
3. Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1/Q_2 is proportional to :



(A) $\frac{x_2^2}{x_1^2}$ (B) $\frac{x_1^3}{x_2^3}$ (C) $\frac{x_1}{x_2}$ (D) $\frac{x_2}{x_1}$

PEO325

4. Three charged particle A, B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d. the charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is :

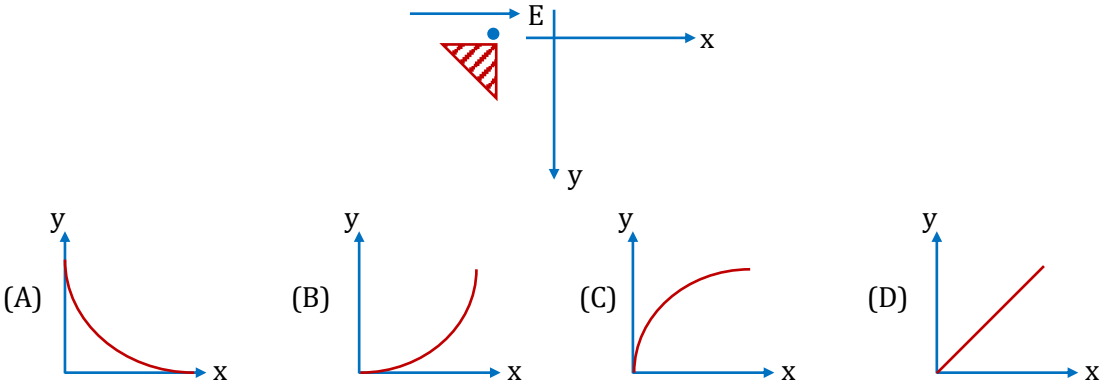


(A) $\frac{2\sqrt{3}\,q}{\pi\varepsilon_0 d^2}$ (B) $\frac{\sqrt{3}\,q}{4\pi\varepsilon_0 d^2}$ (C) $\frac{3\sqrt{3}\,q}{4\pi\varepsilon_0 d^2}$ (D) $\frac{\sqrt{3}\,q}{\pi\varepsilon_0 d^2}$

PEO326

For More Join: @IITJEE_Advanced

5. A small point mass carrying some positive charge on it, is released from the edge of a table. There is a uniform electric field in this region in the horizontal direction. Which of the following options then correctly describe the trajectory of the mass ? (Curves are drawn schematically and are not to scale).

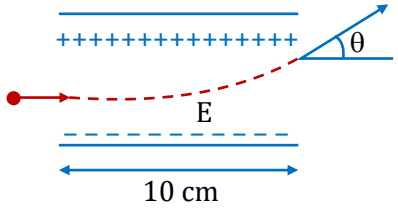


PEO327

6. A stream of a positively charged particles having $\frac{q}{m}=2\times10^{11}\frac{C}{kg}$ and velocity $\vec{v}_0=3\times10^7\hat{i}\ m/s$ is deflected by an electric field $1.8\hat{j}k\ V/m$. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charge particles in the y-direction is _____mm.

PEO328

7. A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV. The length of each plate is 10 cm. The angle (θ) of deviation of the path of electron as it comes out of the field is_____ (in degree).



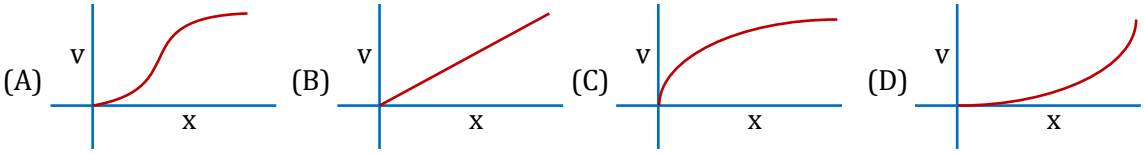
PEO329

8. The bob of a simple pendulum has mass 2g and a charge of 5.0 μC. It is at rest in a uniform horizontal electric field of intensity 2000 V/m. At equilibrium, the angle that the pendulum makes with the vertical is: (take g = 10 m/s^2)

(A) $\tan^{-1}(5.0)$ (B) $\tan^{-1}(2.0)$ (C) $\tan^{-1}(0.5)$ (D) $\tan^{-1}(0.2)$

PEO330

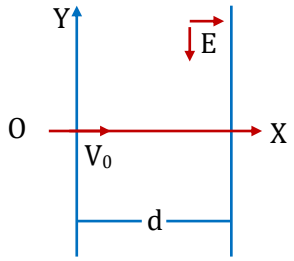
9. A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale)



PEO331

For More Join: @IITJEE_Advanced

10. A charged particle (mass m and charge q) moves along X axis with velocity V_0. When it passes through the origin it enters a region having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is :



(A) $y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$

(B) $y = \frac{qEd}{mV_0^2}(x - d)$

(C) $y = \frac{qEd}{mV_0^2}x$

(D) $y = \frac{qEd^2}{mV_0^2}x$

PEO332

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	C	D	D	2	45	C	C	A

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Electrostatics : RACE-03

1. For a uniformly charged ring of radius R, the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is :

(A) $\frac{R}{\sqrt{5}}$ (B) R (C) $\frac{R}{\sqrt{2}}$ (D) $R\sqrt{2}$

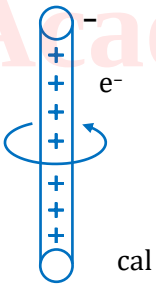
PEO333

2. Suppose a uniformly charged wall provides a uniform electric field of $2 \times 10^4\ N/C$ normally. A charged particle of mass $2\ g$ being suspended through a silk thread of length $20\ cm$ and remain stayed at a distance of $10\ cm$ from the wall. Then the charge on the particle will be $\frac{1}{\sqrt{x}}\mu C$ where x =_____. [use $g = 10\ m/s^2$]

PEO334

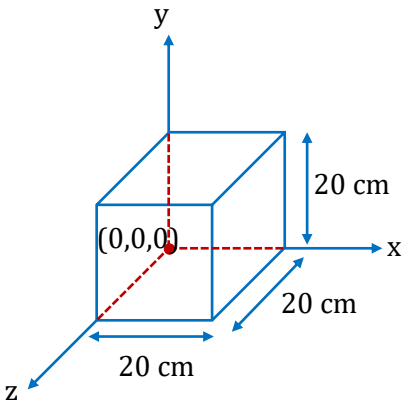
3. An electron revolves around an infinite cylindrical wire having uniform linear charge density $2 \times 10^{-8}\ Cm^{-1}$ in circular path under the influence of attractive electrostatic field as shown in the figure. The velocity of electron with which it is revolving is_____$\times 10^6\ ms^{-1}$. Given mass of electron $= 9 \times 10^{-31}\ kg$.

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PEO335

4. Expression for an electric field is given by $\vec{E} = 4000x^2\hat{\imath}\frac{V}{m}$. The electric flux through the cube of side $20\ cm$ when placed in electric field (as shown in the figure) is _____$V\ cm$.



PEO336

For More Join: @IITJEE_Advanced

5. An electric field is given by $(6\hat{\imath} + 5\hat{\jmath} + 3\hat{k})\ N/C$. The electric flux through a surface area $30\hat{\imath}\ m^2$ lying in *YZ*-plane (in *SI* unit) is :

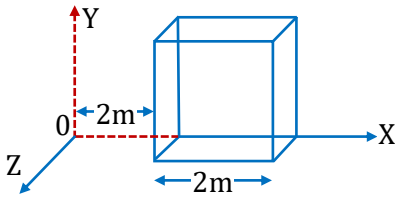
(A) 90 (B) 150 (C) 180 (D) 60

PEO337

6. An electric field, $\vec{E} = \frac{2\hat{\imath}+6\hat{\jmath}+8\hat{k}}{\sqrt{6}}$ passes through the surface of 4 m^2 area having unit vector $\hat{n} = \left(\frac{2\hat{\imath}+\hat{\jmath}+\hat{k}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ $V\,m$.

PEO338

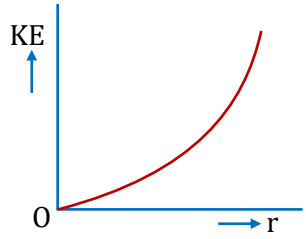
7. An electric field $\vec{E} = (2x\hat{\imath})NC^{-1}$ exists in space. A cube of side $2m$ is placed in the space as per figure given below. The electric flux through the cube is ..... Nm^2/C.

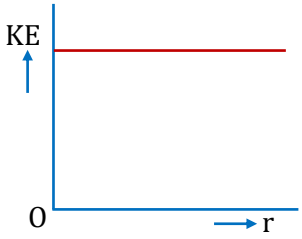


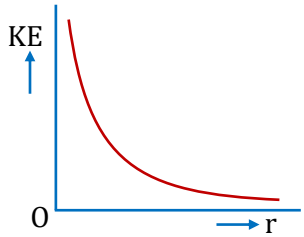
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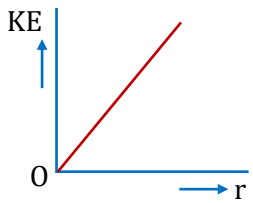
PEO339

8. An infinitely long positively charged straight thread has a linear charge density $\lambda\ Cm^{-1}$. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :

(A)


(B)


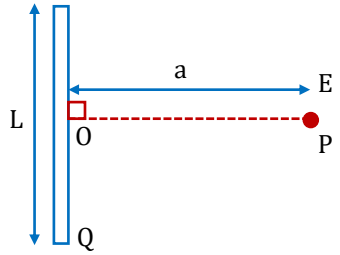
(C)


(D)


PEO340

For More Join: @IITJEE_Advanced

9. Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the rod is $a=\frac{\sqrt{3}}{2}L$.



(A) $\frac{\sqrt{3}Q}{4\pi\varepsilon_0 L^2}$ (B) $\frac{Q}{3\pi\varepsilon_0 L^2}$ (C) $\frac{Q}{2\sqrt{3}\pi\varepsilon_0 L^2}$ (D) $\frac{Q}{4\pi\varepsilon_0 L^2}$

PEO341

10. A particle of charge '$-q$' and mass 'm' moves in a circle of radius 'r' around an infinitely long line charge of linear density '$+\lambda$'. Then time period will be given as:
(Consider k as Coulomb's constant)

(A) $T^2=\frac{4\pi^2 m}{2k\lambda q}r^3$ (B) $T=2\pi r\sqrt{\frac{m}{2k\lambda q}}$ (C) $T=\frac{1}{2\pi r}\sqrt{\frac{m}{2k\lambda q}}$ (D) $T=\frac{1}{2\pi}\sqrt{\frac{2k\lambda q}{m}}$

PEO342

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	3	8	640	C	12	16	B	C	B

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Electrostatics : RACE-04

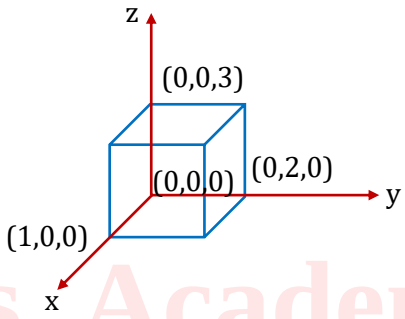
1. In a cuboid of dimension $2L \times 2L \times L$, a charge q is placed at the centre of the surface 'S' having area of 4 L^2. The flux through the opposite surface to 'S' is given by

(A) $\frac{q}{12\varepsilon_0}$ (B) $\frac{q}{3\varepsilon_0}$ (C) $\frac{q}{2\varepsilon_0}$ (D) $\frac{q}{6\varepsilon_0}$

PEO343

2. As shown in figure, a cuboid lies in a region with electric field $E = 2x^2\hat{\imath} - 4y\hat{\jmath} + 6\hat{k}$ N/C. The magnitude of charge within the cuboid is $n \in_0 C$. The value of n is _____.

(if dimension of cuboid is 1 × 2 × 3 m^3)



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PEO344

3. A cubical volume is bounded by the surfaces $x = 0, x = a, y = 0, y = a, z = 0, z = a$. The electric field in the region is given by $\vec{E} = E_0 x \hat{\imath}$. Where $E_0 = 4 \times 10^4\ NC^{-1}\ m^{-1}$. If $a = 2\ cm$, the charge contained in the cubical volume is $Q \times 10^{-14} C$. The value of Q is___. (Take: $\in_0 = 9 \times 10^{-12}\ C^2/Nm^2$)

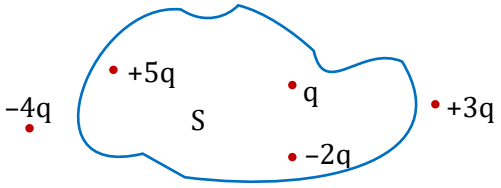
PEO345

4. A charge q is placed at the center of one of the surface of a cube. The flux linked with the cube is :-

(A) $\frac{q}{4\in_0}$ (B) $\frac{q}{2\in_0}$ (C) $\frac{q}{8\in_0}$ (D) Zero

PEO346

5. Five charges $+q$, $+5q$, $-2q$, $+3q$ and $-4q$ are situated as shown in the figure. The electric flux due to this configuration through the surface S is :



(A) $\frac{5q}{\in_0}$ (B) $\frac{4q}{\in_0}$ (C) $\frac{3q}{\in_0}$ (D) $\frac{q}{\in_0}$

PEO347

For More Join: @IITJEE_Advanced

6. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0\left(\frac{5}{4}-\frac{r}{R}\right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance r ($r < R$) from the origin is given by :

(A) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$ (B) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$ (C) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$ (D) $\frac{4\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$

PEO348

7. Let a total charge $2Q$ be distributed in a sphere of radius R, with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B, of $-Q$ each, are placed on diametrically opposite points, at equal distance, a, from the centre. If A and B do not experience any force, then:

(A) $a = \frac{3R}{2^{1/4}}$ (B) $a = R/\sqrt{3}$ (C) $a = 8^{-1/4}R$ (D) $a = 2^{-1/4}R$

PEO349

8. In finding the electric field using Gauss Law the formula $|\vec{E}| = \frac{q_{enc}}{\varepsilon_0 |A|}$ is applicable. In the formula ε_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. The equation can be used in which of the following situation?

(A) Only when the Gaussian surface is an equipotential surface.

(B) Only when $|\vec{E}|$ = constant on the surface.

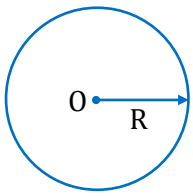
(C) For any choice of Gaussian surface.

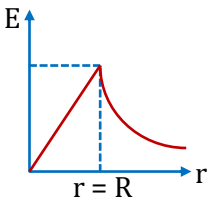
(D) Only when the Gaussian surface is an equipotential surface and $|\vec{E}|$ is constant on the surface.

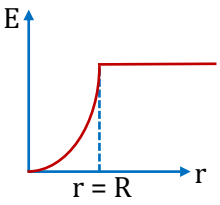
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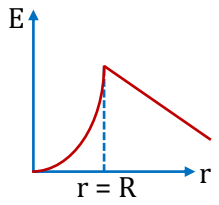
PEO350

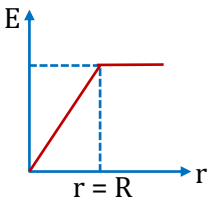
9. Graphical variation of electric field due to a uniformly charged insulating solid sphere of radius R, with distance r from the centre O is represented by:



(A) 

(B) 

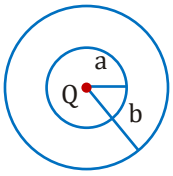
(C) 

(D) 

PEO351

For More Join: @IITJEE_Advanced

10. The region between two concentric spheres of radii 'a' and 'b', respectively (see figure), has volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q. The value of A such that the electric field in the region between the spheres will be constant, is:-



(A) $\frac{2Q}{\pi a^2}$ (B) $\frac{Q}{2\pi a^2}$ (C) $\frac{Q}{2\pi(b^2-a^2)}$ (D) $\frac{2Q}{\pi(a^2-b^2)}$

PEO352

Rankers Academy JEE

ANSWER KEY

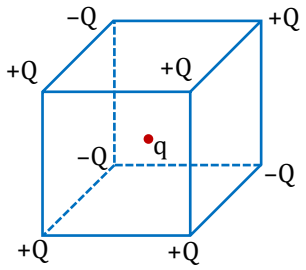
Question	1	2	3	4	5	6	7	8	9	10
Answer	D	12	288	B	B	C	C	D	A	B

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Electrostatics : RACE-05

1. Nine point charges are placed on a cube as shown in the figure. The charge q is placed at the body centre whereas all other charges are at the vertices. The electrostatic potential energy of the system will be :-

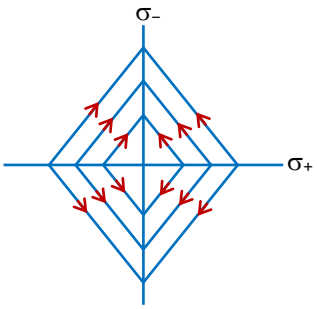


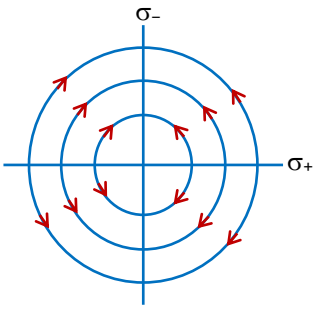
(A) Positive (B) Negative (C) Zero (D) Depends on sign and value of q

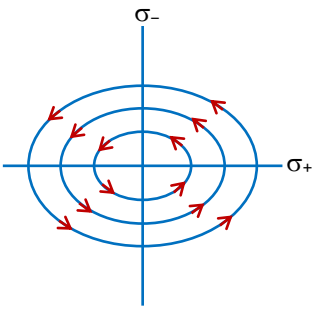
PEO353

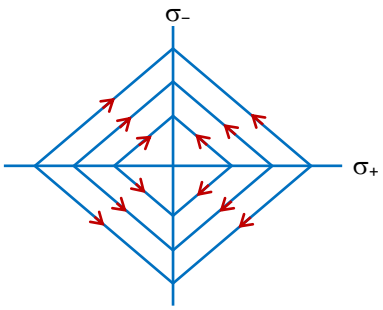
Rankers Academy JEE

2. Two charged thin infinite plane sheets of uniform surface charge density σ_+ and σ_- where $|\sigma_+| > |\sigma_-|$ intersect at right angle. Which of the following best represents the electric field lines for this system :

(A)


(B)


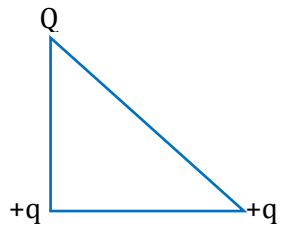
(C)


(D)


PEO354

For More Join: @IITJEE_Advanced

3. The charges Q, $+q$ and $+q$ are placed at the vertices of a right-angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is:



(A) $\frac{-\sqrt{2}q}{\sqrt{2}+1}$ (B) $-2q$ (C) $\frac{-q}{1+\sqrt{2}}$ (D) $+q$

PEO355

4. Four equal point charges Q each are placed in the xy plane at (0, 2), (4, 2), (4, –2) and (0, –2). The work required to put a fifth charge Q at the origin of the coordinate system will be :

(A) $\frac{Q^2}{2\sqrt{2}\pi\varepsilon_0}$ (B) $\frac{Q^2}{4\pi\varepsilon_0}\left(1+\frac{1}{\sqrt{5}}\right)$ (C) $\frac{Q^2}{4\pi\varepsilon_0}\left(1+\frac{1}{\sqrt{3}}\right)$ (D) $\frac{Q^2}{4\pi\varepsilon_0}$

PEO356

5. The electric field in a region is given by $\vec{E} = (Ax + B)\hat{\imath}$, where E is in NC^{-1} and x is in metres. The values of constants are A = 20 SI unit and B = 10 SI unit. If the potential at x = 1 is V_1 and that at x = –5 is V_2, then $V_1 - V_2$ is :-

(A) –48 V (B) –520 V (C) 180 V (D) 320 V

PEO357

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6. A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x-direction, where a and E_0 are constants. Initially the particle was at rest at x = 0. Other than the initial position the kinetic energy of the particle becomes zero when the distance of the particle from the origin is:

(A) $\sqrt{\frac{2}{a}}$ (B) $\sqrt{\frac{1}{a}}$ (C) a (D) $\sqrt{\frac{3}{a}}$

PEO358

7. If the electric potential at any point $(x, y, z)m$ in space is given by $V = 3x^2$ volt. The electric field at the point (1, 0, 3) m will be

(A) 3 Vm^{-1}, directed along positive x-axis. (B) 3 Vm^{-1}, directed along negative x-axis.

(C) 6 Vm^{-1}, directed along positive x-axis. (D) 6 Vm^{-1}, directed along negative x-axis.

PEO359

8. An electric charge $10^{-6}\mu C$ is placed at origin (0, 0) m of $X-Y$ co-ordinate system. Two points P and Q are situated at $(\sqrt{3},\sqrt{3})m$ and $(\sqrt{6},0)m$ respectively. The potential difference between the points P and Q will be :

(A) $\sqrt{3}V$ (B) $\sqrt{6}V$ (C) 0 V (D) 3 V

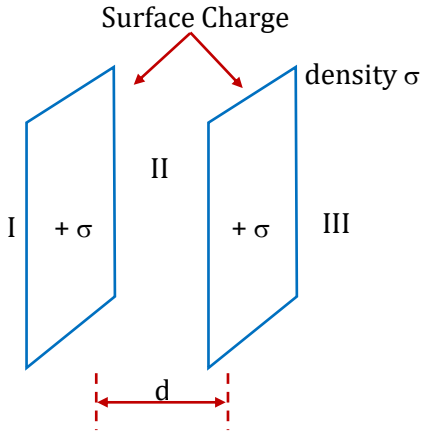
PEO360

9. The electric potential at the surface of an atomic nucleus (z = 50) of radius 9×10^{-13} cm is _____ $\times 10^6$ V.

PEO361

For More Join: @IITJEE_Advanced

10. Let σ be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in three different region E_I, E_{II} and E_{III} are:



(A) $\vec{E}_I = \frac{2\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{2\sigma}{\epsilon_0}\hat{n}$

(B) $\vec{E}_I = 0, \vec{E}_{II} = \frac{\sigma}{\epsilon_0}\hat{n}, \vec{E}_{III} = 0$

(C) $\vec{E}_I = \frac{\sigma}{2\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{2\epsilon_0}\hat{n}$

(D) $\vec{E}_I = -\frac{\sigma}{\epsilon_0}\hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{\epsilon_0}\hat{n}$

PEO362

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	A	A	B	C	D	D	C	8.00	D

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Electrostatics : RACE-06

1. A uniformly charged ring of radius $3a$ and total charge q is placed in xy-plane centred at origin. A point charge q is moving towards the ring along the z-axis and has speed u at $z = 4a$. The minimum value of u such that it crosses the origin is :

(A) $\sqrt{\frac{2}{m}\left(\frac{1}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)}^{1/2}$ (B) $\sqrt{\frac{2}{m}\left(\frac{2}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)}^{1/2}$ (C) $\sqrt{\frac{2}{m}\left(\frac{4}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)}^{1/2}$ (D) $\sqrt{\frac{2}{m}\left(\frac{1}{5}\frac{q^2}{4\pi\varepsilon_0 a}\right)}^{1/2}$

PEO363

2. In free space, a particle A of charge 1 μC is held fixed at a point P. Another particle B of the same charge and mass 4 μg is kept at a distance of 1 mm from P. if B is released, then its velocity at a distance of 9 mm from P is:

$\left[\text{Take } \frac{1}{4\pi\varepsilon_0} = 9\times10^9\, Nm^2\, C^{-2}\right]$

(A) $2.0 \times 10^3\ m/s$ (B) $3.0 \times 10^4\ m/s$ (C) $1.5 \times 10^2\ m/s$ (D) $1.0\ m/s$

PEO364

3. The two thin coaxial rings, each of radius 'a' and having charges $+Q$ and $-Q$ respectively are separated by a distance of 's'. The potential difference between the centres of the two rings is :

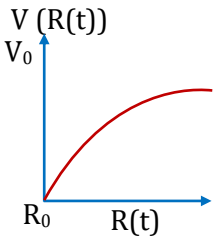
(A) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a}+\frac{1}{\sqrt{s^2+a^2}}\right]$ (B) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a}+\frac{1}{\sqrt{s^2+a^2}}\right]$

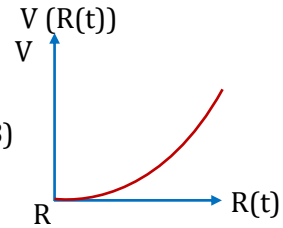
(C) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a}-\frac{1}{\sqrt{s^2+a^2}}\right]$ (D) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a}-\frac{1}{\sqrt{s^2+a^2}}\right]$

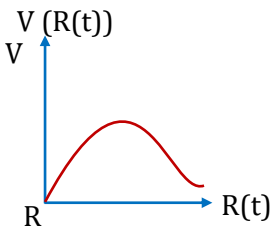
Rankers Academy JEE

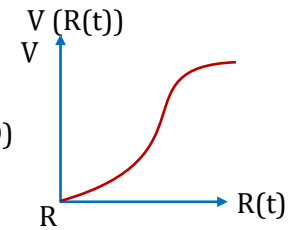
PEO365

4. There is a uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed $V(R(t))$ of the distribution as a function of its instantaneous radius $R(t)$ is :

(A)


(B)


(C)


(D)


PEO366

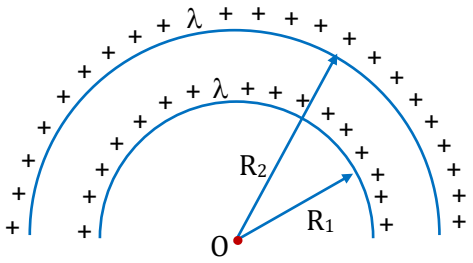
For More Join: @IITJEE_Advanced

5. An electric field $\vec{E}=\left(25\hat{i}+30\hat{j}\right)NC^{-1}$ exists in a region of space. If the potential at the origin is taken to be zero then the potential at $x = 2m, y = 2m$ is :-

(A) $-140\ J$ (B) $-120\ J$ (C) $-110\ J$ (D) $-130\ J$

PEO367

6. The electric potential at the centre of two concentric half rings of radii R_1 and R_2, having same linear charge density λ is



(A) $\frac{2\lambda}{\epsilon_0}$ (B) $\frac{\lambda}{2\epsilon_0}$ (C) $\frac{\lambda}{4\epsilon_0}$ (D) $\frac{\lambda}{\epsilon_0}$

PEO368

7. A uniformly charged solid sphere of radius R has potential V_0 (measured with respect to ∞) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3V_0}{2}, \frac{5V_0}{4}, \frac{3V_0}{4}$ and $\frac{V_0}{4}$ have radius R_1, R_2, R_3 and R_4 respectively. Then

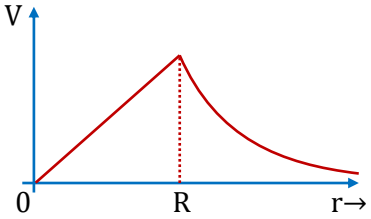
(A) $R_1 = 0$ and $R_2 < (R_4 - R_3)$ (B) $2R < R_4$

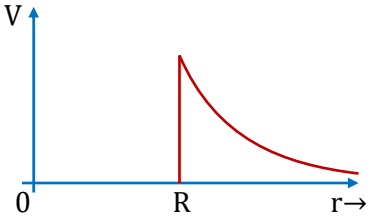
(C) $R_1 = 0$ and $R_2 > (R_4 - R_3)$ (D) $R_1 \neq 0$ and $(R_2 - R_1) > (R_4 - R_3)$

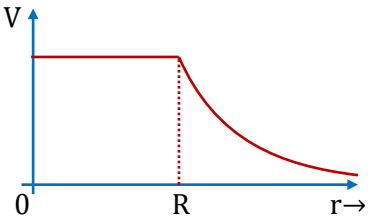
Pankers Academy JEE

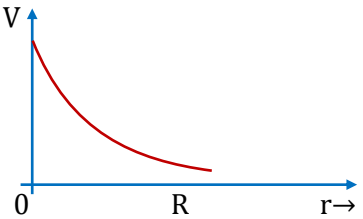
PEO369

8. Which of the following correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the centre ?

(A)


(B)


(C)


(D)


PEO370

9. 64 identical drops each charged upto potential of 10 mV are combined to form a bigger drop. The potential of the bigger drop will be _____ mV.

PEO371

For More Join: @IITJEE_Advanced

10. Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2, respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1 (on S_1) / V_2 (on S_2) of the electrostatic potentials on each sphere is :

(A) (R_2/R_1) (B) $\left(\frac{R_1}{R_2}\right)^3$ (C) R_1/R_2 (D) $(R_1/R_2)^2$

PEO372

Rankers Academy JEE

ANSWER KEY

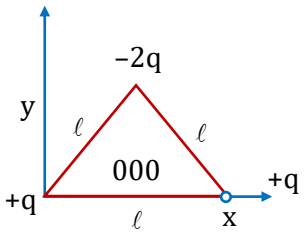
Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	D	A	C	B	A or B	C	160	D

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Electrostatics : RACE-07

1. Determine the electric dipole moment of the system of three charges, placed on the vertices of an equilateral triangle, as shown in the figure:



(A) $(q\ell)\dfrac{\hat{i}+\hat{j}}{\sqrt{2}}$ (B) $\sqrt{3}q\ell\dfrac{\hat{j}-\hat{i}}{\sqrt{2}}$ (C) $-\sqrt{3}q\,\ell\,\hat{j}$ (D) $2q\ell\,\hat{j}$

PEO373

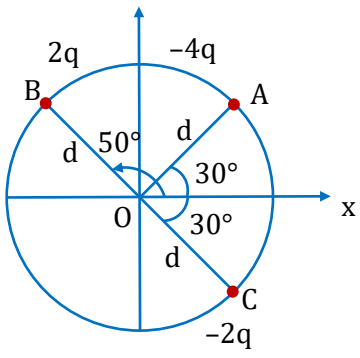
2. A point dipole $\vec{p}=-p_0\hat{x}$ is kept at the origin. The potential and electric field due to this dipole on the y-axis at a distance d are, respectively:
(Take $V = 0$ at infinity) :

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(A) $\dfrac{|\vec{p}|}{4\pi\varepsilon_0 d^2}, \dfrac{-\vec{p}}{4\pi\varepsilon_0 d^3}$ (B) $0, \dfrac{\vec{p}}{4\pi\varepsilon_0 d^3}$ (C) $\dfrac{|\vec{p}|}{4\pi\varepsilon_0 d^2}, \dfrac{\vec{p}}{4\pi\varepsilon_0 d^3}$ (D) $0, \dfrac{-\vec{p}}{4\pi\varepsilon_0 d^3}$

PEO374

3. Three charged particle A, B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d. the charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is :

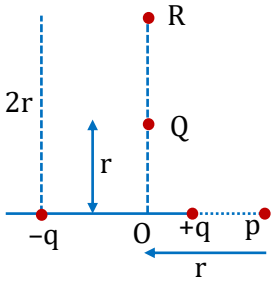


(A) $\dfrac{2\sqrt{3}\,q}{\pi\varepsilon_0 d^2}$ (B) $\dfrac{\sqrt{3}\,q}{4\pi\varepsilon_0 d^2}$ (C) $\dfrac{3\sqrt{3}\,q}{4\pi\varepsilon_0 d^2}$ (D) $\dfrac{\sqrt{3}\,q}{\pi\varepsilon_0 d^2}$

PEO375

For More Join: @IITJEE_Advanced

4. The electric field at point p due to an electric dipole is E. The electric field at point R on equitorial line will be $\frac{E}{x}$. The value of x :



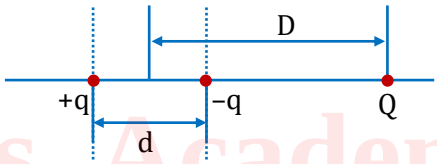
PEO376

5. Four-point charges $-q$, $+q$, $+q$ and $-q$ are placed on y-axis at $y = -2d$, $y = -d$, $y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x-axis at $x = D$, with $D >> d$, will behave as :-

(A) $E \propto \frac{1}{D}$ (B) $E \propto \frac{1}{D^3}$ (C) $E \propto \frac{1}{D^2}$ (D) $E \propto \frac{1}{D^4}$

PEO377

6. A system of three charges are placed as shown in the figure :



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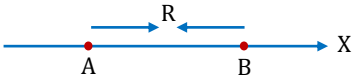
If $D >> d$, the potential energy of the system is best given by :

(A) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}-\frac{qQd}{2D^2}\right]$ (B) $\frac{1}{4\pi\varepsilon_0}\left[+\frac{q^2}{d}+\frac{qQd}{D^2}\right]$

(C) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}+\frac{2qQd}{D^2}\right]$ (D) $\frac{1}{4\pi\varepsilon_0}\left[-\frac{q^2}{d}-\frac{qQd}{D^2}\right]$

PEO378

7. Two electric dipoles, A, B with respective dipole moments $\vec{d}_A = -4qa\hat{i}$ and $\vec{d}_B = -2qa\hat{i}$ placed on the x-axis with a separation R, as shown in the figure:



The distance from A at which both of them produce the same potential is :

(A) $\frac{\sqrt{2}R}{\sqrt{2}+1}$ (B) $\frac{R}{\sqrt{2}+1}$ (C) $\frac{\sqrt{2}\,R}{\sqrt{2}-1}$ (D) $\frac{R}{\sqrt{2}-1}$

PEO379

8. An electric dipole of moment $\vec{p} = (-\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29} C.m$ is at the origin (0, 0, 0). The electric field due to this dipole at $\vec{r} = +\hat{i} + 3\hat{j} + 5\hat{k}$ (note that $\vec{r}.\vec{p} = 0$) is parallel to:

(A) $(-\hat{i} + 3\hat{j} - 2\hat{k})$ (B) $(+\hat{i} - 3\hat{j} - 2\hat{k})$ (C) $(+\hat{i} + 3\hat{j} - 2\hat{k})$ (D) $(-\hat{i} - 3\hat{j} + 2\hat{k})$

PEO380

For More Join: @IITJEE_Advanced

9. An electric dipole is formed by two equal and opposite charges q with separation d. The charges have same mass m. It is kept in a uniform electric field E. If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is :-

(A) $\sqrt{\frac{qE}{2md}}$ (B) $2\sqrt{\frac{qE}{md}}$ (C) $\sqrt{\frac{2qE}{md}}$ (D) $\sqrt{\frac{qE}{md}}$

PEO381

10. An electric dipole of dipole moment is $6.0\times10^{-6}Cm$ placed in a uniform electric field of $1.5\times10^{3}NC^{-1}$ in such a way that dipole moment is along electric field. The work done in rotating dipole by 180° in this field will be _____mJ.

PEO382

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	D	D	16	D	D	C	C	C	18

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Electrostatics : RACE-08

1. Given below are two statements:

Statement I : Electric potential is constant within and at the surface of each conductor.

Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.

In the light of the above statements, choose the most appropriate answer from the options give below.

(A) Both statement I and statement II are correct.

(B) Both statement I and statement II are incorrect.

(C) Statement I is correct but statement II is incorrect.

(D) Statement I is incorrect but and statement II is correct.

PEO383

2. A charge Q is distributed over three concentric spherical shells of radii a, b, c ($a < b < c$) such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where $r < a$, would be:

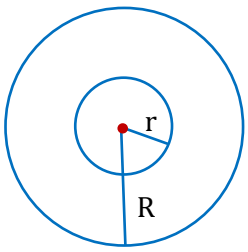
(A) $\frac{Q}{4\pi\varepsilon_0(a+b+c)}$ (B) $\frac{Q(a+b+c)}{4\pi\varepsilon_0(a^2+b^2+c^2)}$

(C) $\frac{Q}{12\pi\varepsilon_0}\frac{ab+bc+ca}{abc}$ (D) $\frac{Q}{4\pi\varepsilon_0}\frac{(a^2+b^2+c^2)}{(a^3+b^3+c^3)}$

Rankers Academy JEE

PEO384

3. A charge Q is distributed over two concentric conducting thin spherical shells radii r and R ($R > r$). If the surface charge densities on the two shells are equal, the electric potential at the common centre is :



(A) $\frac{1}{4\pi\varepsilon_0}\frac{(R+2r)Q}{2(R^2+r^2)}$ (B) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)}{2(R^2+r^2)}Q$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{(R+r)}{(R^2+r^2)}Q$ (D) $\frac{1}{4\pi\varepsilon_0}\frac{(2R+r)}{(R^2+r^2)}Q$

PEO385

4. Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have 12 μC and –3 μC charges, respectively, and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively :

(A) 6 μC and 3 μC (B) +4.5 μC and –4.5 μC

(C) 3 μC and 6 μC (D) 4.5 μC on both

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5. Consider the force F on a charge 'q' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F, if 'q' is placed at distance r from the centre of the shell ?

(A) $F=\frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}$ for $r>R$ (B) $\frac{1}{4\pi\varepsilon_0}\frac{qQ}{R^2}>F>0$ for $r<R$

(C) $F=\frac{1}{4\pi\varepsilon_0}\frac{Qq}{r^2}$ for all r (D) $F=\frac{1}{4\pi\varepsilon_0}\frac{Qq}{R^2}$ for $r<R$

PEO387

6. σ is the uniform surface charge density of a thin spherical shell of radius R. The electric field at any point on the surface of the spherical shell is :

(A) $\sigma/\in_0 R$ (B) $\sigma/2\in_0$ (C) $\sigma/\in_0$ (D) $\sigma/4\in_0$

PEO388

7. The vehicles carrying inflammable fluids usually have metallic chains touching the ground :

(A) To conduct excess charge due to air friction to ground and prevent sparking.

(B) To alert other vehicles.

(C) To protect tyres from catching dirt from ground.

(D) It is a custom.

PEO389

8. Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is:

(A) $\sqrt{ab}$ (B) ab (C) $\frac{a}{b}$ (D) $\frac{b}{a}$

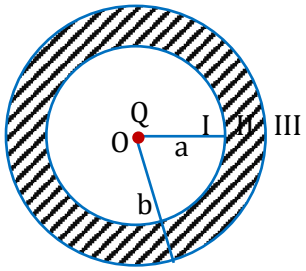
PEO390

9. Two isolated metallic solid spheres of radii R and $2R$ are charged such that both have same charge density σ. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ'. The ratio $\frac{\sigma'}{\sigma}$ is :

(A) $\frac{9}{4}$ (B) $\frac{4}{3}$ (C) $\frac{5}{3}$ (D) $\frac{5}{6}$

PEO391

10. As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b. The electric field due to charge Q in three different regions I, II and III is given by : ($I : r<a$, $II : a<r<b$, $III : r>b$)



(A) $E_I = 0, E_{II} = 0, E_{III} \neq 0$ (B) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$

(C) $E_I \neq 0, E_{II} = 0, E_{III} = 0$ (D) $E_I = 0, E_{II} = 0, E_{III} = 0$

PEO392

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	C	A	A	C	A	C	D	B

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